

Learning Note: Editing Record

Updated 7 October 2026

Purpose

Record substantive changes between editing sessions. This is a dated human-readable record, without numbered releases, approval procedures, or duplicate versions. Future edits append a new dated entry and regenerate this PDF.

2 October 2026 — Initial note

Created: `Electromagnetism-Foundations/main.tex` and its PDF, plus independent learning-plan and editing-record documents in `Documents/`.

Content: Operational definitions and experimental access to electric and magnetic fields; magnetic moments, electron spin, stationary magnets, and energy; definitions of flux, divergence, and curl; fluid, heat, current, and electromagnetic applications; illustrative calculations; and a limited map of related physical laws.

Clarifications from the supplied discussion:

- Electric field is external force per signed test charge in a specified frame, with negligible probe disturbance.
- The magnetic force is perpendicular to velocity and field; the field need not be perpendicular to velocity. One velocity direction cannot determine every field component.
- Spin is intrinsic angular momentum, not literal mechanical rotation. Atomic beam splitting is distinguished from a free-electron experiment.
- Fixed-dipole torque, force, and interaction-energy formulas state their approximation and distinguish uniform from nonuniform external fields.
- Static magnetic fields and intrinsic spin do not require continuous power; resistive coils have losses.
- Field flux is not necessarily material transport. Zero divergence does not mean zero field. Circular field lines do not necessarily mean nonzero local curl.
- Definitions and mathematical theorems are distinguished from physical laws. The history is not reduced to a universal definitions-first sequence.

Build and organization: Self-contained pdfLaTeX source with embedded references and diagram, suitable for Overleaf. Stable filenames, one topic folder, and only TeX/PDF document files intended for Git. Build artifacts are excluded locally.

Validation: Compiled locally with pdfLaTeX, resolved citations on a second pass, and inspected rendered PDF pages. Recompilation does not require external images, BibTeX, or shell escape.

Next: Begin Maxwell's equations only when requested, following the learning plan. Wave propagation, reflection, and detailed AN720 analysis remain future lessons.

6 October 2026 — Laws, experiments, and timeline

New note: Added Electromagnetism-Foundations/

laws-experiments-and-timeline.tex and its PDF as an independent lesson alongside the original note.

Content: Historical organization of Coulomb, steady-current magnetism, Gauss's law, Faraday, Lenz, Stokes, Maxwell's extension, and the combined force law. Definitions, experimentally established laws, mathematical theorems, and derived special cases are identified separately.

Clarifications: Expanded the independent electrical and magnetic measurements behind Faraday's equality. Explained EMF's voltage units, the tangential directions in $\mathbf{E} \cdot d\boldsymbol{\ell}$, boundary versus spanning surface, orientation and signs, and how changing magnetic flux relates to electric circulation. The stationary-loop restriction, total versus imposed magnetic field, receiving-circuit resistance/inductance, and proof versus experimental evidence are explicit.

Rename: Renamed the original main.tex/main.pdf pair to fields-and-vector-calculus.tex and its PDF; retained its content and updated compilation instructions and companion-note reference.

Plan: Recorded the new lesson without treating it as confirmed mastery. Added a suggested reading path from physical foundations through mathematical electrodynamics, transmission lines/RF, and antenna design.

Build: Both topic notes remain self-contained pdfLaTeX sources with embedded bibliography and diagrams. Only TeX/PDF files belong in Git; auxiliary files are generated outside the repository. PDF validation is performed by local compilation and rendered-page inspection.

7 October 2026 — Shift to book-led study

Primary text: The learning sequence now follows Edward M. Purcell and David J. Morin, *Electricity and Magnetism*, 3rd edition (2013). The two existing foundations notes serve as references alongside the book.

Reading checkpoint: Chapter 1, *Electrostatics: charges and fields*. Discussion has covered Coulomb's law, superposition, Gauss's law and its geometric derivation, fields of spherical/line/sheet charge distributions, forces on surface charge, and field energy. Problem 1.33 was worked through to verify the interaction-energy term for two charges. These entries record subjects discussed, not a claim that every subject has been mastered.

Plan: Replaced the immediate Maxwell-first step with book-led study beginning with electrostatics, followed by potential and conductors. Retained the later goals of physical transmission-line reflection and the AN720 antenna. Physical interpretation, explicit assumptions, and complete derivations remain the learning priorities; advancement requires the learner's confirmation.

Documents: Updated `Documents/plan.tex` and this editing record, and regenerated their PDFs with local pdfLaTeX. This editing session also commits the pending foundations-note rename and the separate laws-and-experiments timeline recorded on 6 October.